

PRACTICAL 7

Aim : Subnet the IP address 216.21.5.0 / 24 into 30 hosts in each subnet.

Input :

1. Given information

- IP address: 216.21.5.0
- Original prefix: /24
- Original subnet mask: 255.255.255.0
- Required hosts per subnet: 30 usable hosts

We need to find the new subnet mask and list all the resulting subnets.

2. Determine how many host bits are required

The formula for the number of usable hosts in a subnet is:

$$2^h - 2$$

where h = number of host bits.

We need at least 30 hosts.

Try 4 host bits:

$$2^4 - 2 = 16 - 2 = 14$$

Only 14 hosts — not enough.

Try 5 host bits:

$$2^5 - 2 = 32 - 2 = 30$$

Exactly 30 usable hosts.

Therefore, we need 5 host bits.

3. Find the new prefix length

The original network is /24.

IPv4 has 32 bits:

$$32 - 24 = 8$$

So the original /24 network has 8 host bits.

We need 5 host bits for each subnet. Therefore, we borrow:

$$8 - 5 = 3 \text{ bits}$$

from the host portion.

So:

$$24 + 3 = \boxed{/27}$$

New subnet prefix

$$\boxed{/27}$$

4. Find the subnet mask

A /27 means the first 27 bits are network bits.

Binary subnet mask:

11111111.11111111.11111111.11100000

Convert each octet to decimal:

$$11111111 = 255$$

$$11111111 = 255$$

$$11111111 = 255$$

$$11100000 = 224$$

Therefore:

$$\boxed{255.255.255.224}$$

So the answer is:

Subnet mask = 255.255.255.224 (/27)

5. Determine the number of subnets

We borrowed 3 bits from the original host portion.

Number of subnets:

$$2^3 = 8$$

Therefore, we get:

8 subnets

Each subnet has:

$$2^5 = 32$$

total IP addresses.

Of these:

- 1 = Network address
- 30 = Usable host addresses
- 1 = Broadcast address

Therefore:

$$32 - 2 = 30 \text{ usable hosts}$$

6. Find the block size

The interesting octet is the fourth octet.

Our subnet mask is:

255.255.255.224

Block size is:

$$256 - 224 = 32$$

So the networks increase by 32 in the fourth octet.

Starting from 0:

0

32

64

96

128

160

192

224

These are our eight network addresses.

7. List all subnets

Subnet 1

Network address:

216.21.5.0

The next network starts at .32, so usable hosts are:

216.21.5.1 – 216.21.5.30

Broadcast:

216.21.5.31

Therefore:

Item	Address
Network	216.21.5.0/27
First host	216.21.5.1
Last host	216.21.5.30
Broadcast	216.21.5.31

Subnet 2

Network address:

216.21.5.32

Usable hosts:

216.21.5.33 – 216.21.5.62

Broadcast:

216.21.5.63

Item	Address
Network	216.21.5.32/27
First host	216.21.5.33
Last host	216.21.5.62
Broadcast	216.21.5.63

Subnet 3

Network address:

216.21.5.64

Usable hosts:

216.21.5.65 – 216.21.5.94

Broadcast:

216.21.5.95

Item	Address
Network	216.21.5.64/27
First host	216.21.5.65
Last host	216.21.5.94
Broadcast	216.21.5.95

Subnet 4

Network address:

216.21.5.96

Usable hosts:

216.21.5.97 – 216.21.5.126

Broadcast:

216.21.5.127

Item	Address
Network	216.21.5.96/27
First host	216.21.5.97
Last host	216.21.5.126
Broadcast	216.21.5.127

Subnet 5

Network address:

216.21.5.128

Usable hosts:

216.21.5.129 – 216.21.5.158

Broadcast:

216.21.5.159

Item	Address
Network	216.21.5.128/27
First host	216.21.5.129
Last host	216.21.5.158
Broadcast	216.21.5.159

Subnet 6

Network address:

216.21.5.160

Usable hosts:

216.21.5.161 – 216.21.5.190

Broadcast:

216.21.5.191

Item	Address
Network	216.21.5.160/27
First host	216.21.5.161
Last host	216.21.5.190
Broadcast	216.21.5.191

Subnet 7

Network address:

216.21.5.192

Usable hosts:

216.21.5.193 – 216.21.5.222

Broadcast:

216.21.5.223

Item	Address
Network	216.21.5.192/27
First host	216.21.5.193
Last host	216.21.5.222
Broadcast	216.21.5.223

Subnet 8

Network address:

216.21.5.224

Usable hosts:

216.21.5.225 – 216.21.5.254

Broadcast:

216.21.5.255

Item	Address
Network	216.21.5.224/27
First host	216.21.5.225
Last host	216.21.5.254
Broadcast	216.21.5.255

8. Complete answer in one table

	Network Address	First Host	Last Host	Broadcast
1	216.21.5.0/27	216.21.5.1	216.21.5.30	216.21.5.31
2	216.21.5.32/27	216.21.5.33	216.21.5.62	216.21.5.63
3	216.21.5.64/27	216.21.5.65	216.21.5.94	216.21.5.95
4	216.21.5.96/27	216.21.5.97	216.21.5.126	216.21.5.127
5	216.21.5.128/27	216.21.5.129	216.21.5.158	216.21.5.159
6	216.21.5.160/27	216.21.5.161	216.21.5.190	216.21.5.191
7	216.21.5.192/27	216.21.5.193	216.21.5.222	216.21.5.223
8	216.21.5.224/27	216.21.5.225	216.21.5.254	216.21.5.255

Final result

Original network: 216.21.5.0/24

New subnet: /27

Subnet mask: 255.255.255.224

Number of subnets: 8

Total addresses per subnet: 32

Usable hosts per subnet: 30

Quick trick to remember

For a /27:

Mask = 255.255.255.224 → Block size = 32

Then simply count by 32:

0, 32, 64, 96, 128, 160, 192, 224

For each one:

Network → Network + 1 through Network + 30 → Network + 31 (Broadcast)